

Iris Classification Using Machine Learning

Internship Project Report

Submitted By

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Domain

Machine Learning Internship

Project Title

Iris Classification Using Logistic Regression

1. Objective

The objective of this project is to classify Iris flowers into different species using machine learning techniques. The Iris dataset contains measurements of flower characteristics such as sepal length, sepal width, petal length, and petal width. Logistic Regression is used to build a classification model capable of predicting the species of a flower based on these features.

2. Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Joblib

3. Dataset Description

The Iris dataset is one of the most popular datasets used in machine learning.

Dataset Information

- Total Records: 150
- Number of Features: 4
- Number of Classes: 3

Features

1. Sepal Length (cm)
2. Sepal Width (cm)
3. Petal Length (cm)
4. Petal Width (cm)

Target Classes

- Iris Setosa
- Iris Versicolor
- Iris Virginica

4. Exploratory Data Analysis (EDA)

The dataset was analyzed before model training.

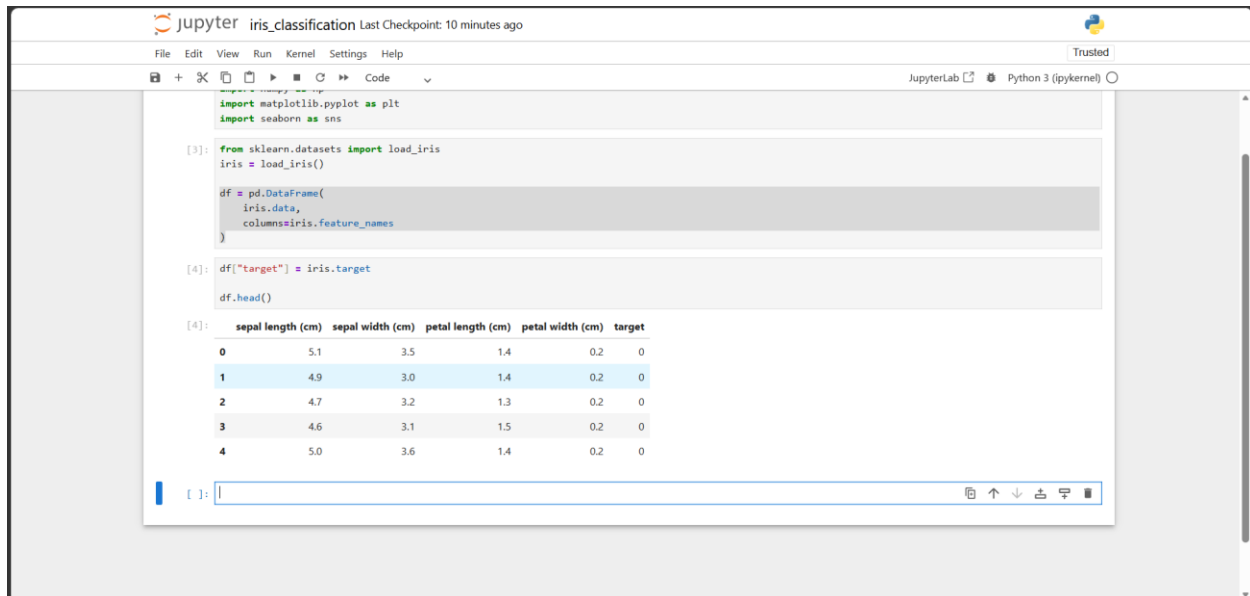
EDA Operations Performed

- Displayed first five rows using `head()`
- Checked dataset information using `info()`
- Generated statistical summary using `describe()`
- Checked missing values using `isnull().sum()`

Observations

- Dataset contains 150 records.
- No missing values were found.
- All features are numerical.
- Data is clean and suitable for machine learning.

Screenshots



JupyterLab interface showing the first three code cells of an `iris_classification` notebook. The notebook title is `iris_classification` and the last checkpoint was 10 minutes ago. The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations, running code, and viewing output. The code cells are as follows:

```
[1]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
[3]: from sklearn.datasets import load_iris
iris = load_iris()

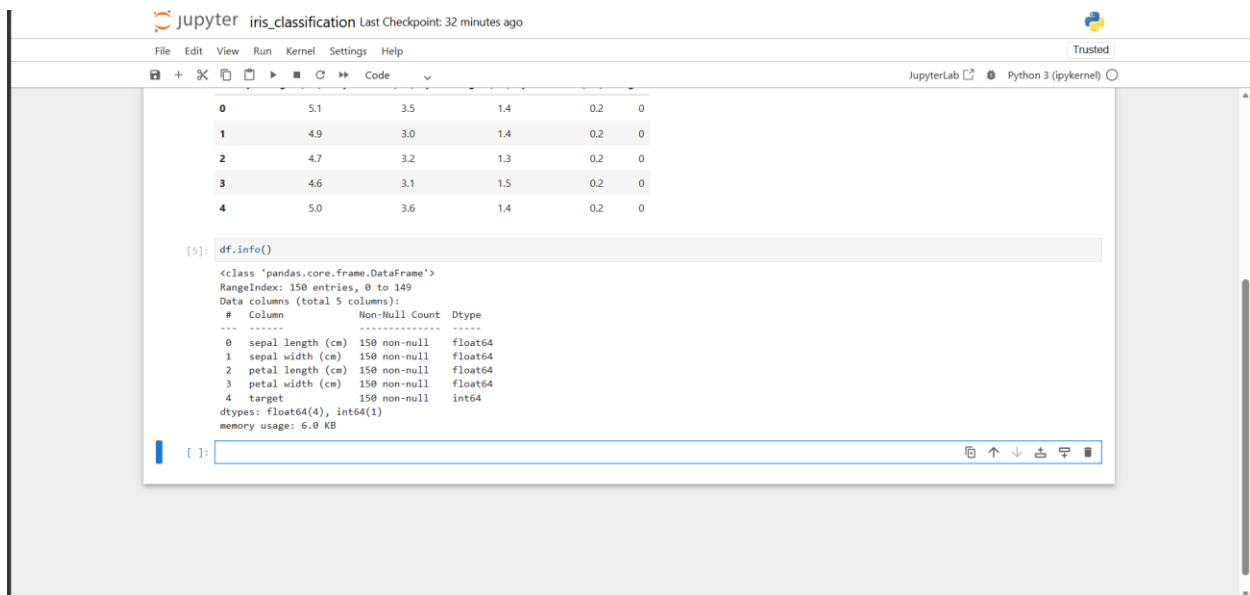
df = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)
```

```
[4]: df["target"] = iris.target

df.head()
```

The output of the third cell shows the first five rows of the DataFrame:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0



JupyterLab interface showing the fourth code cell of the `iris_classification` notebook. The notebook title is `iris_classification` and the last checkpoint was 32 minutes ago. The code cell is as follows:

```
[5]: df.info()
```

The output of the fourth cell shows the summary statistics of the DataFrame:

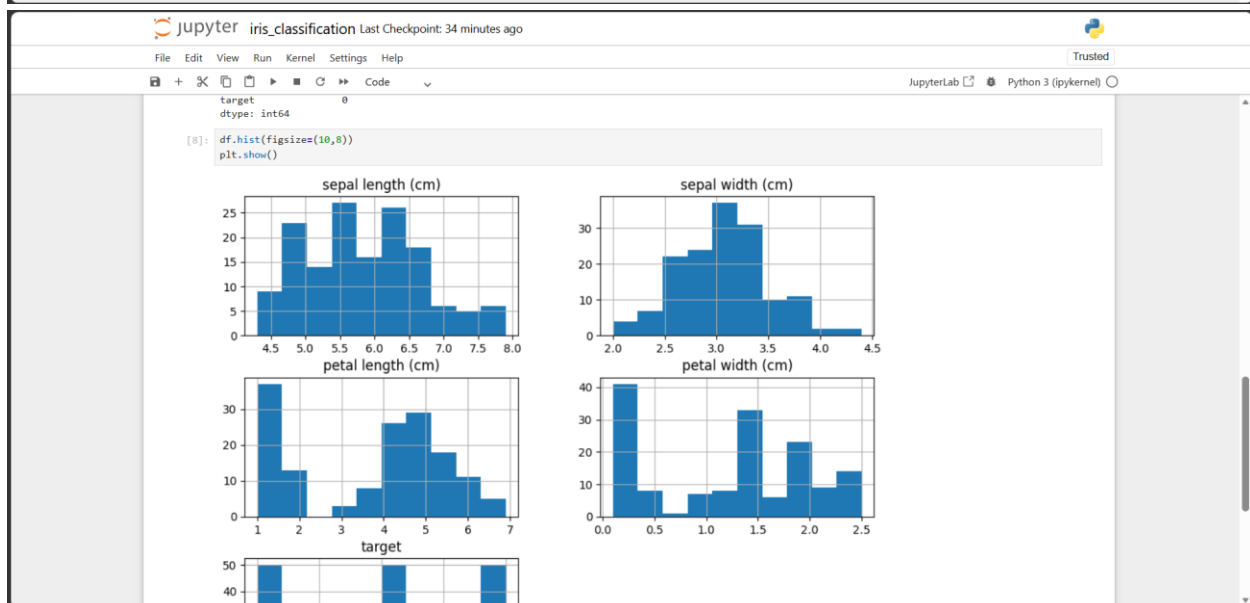
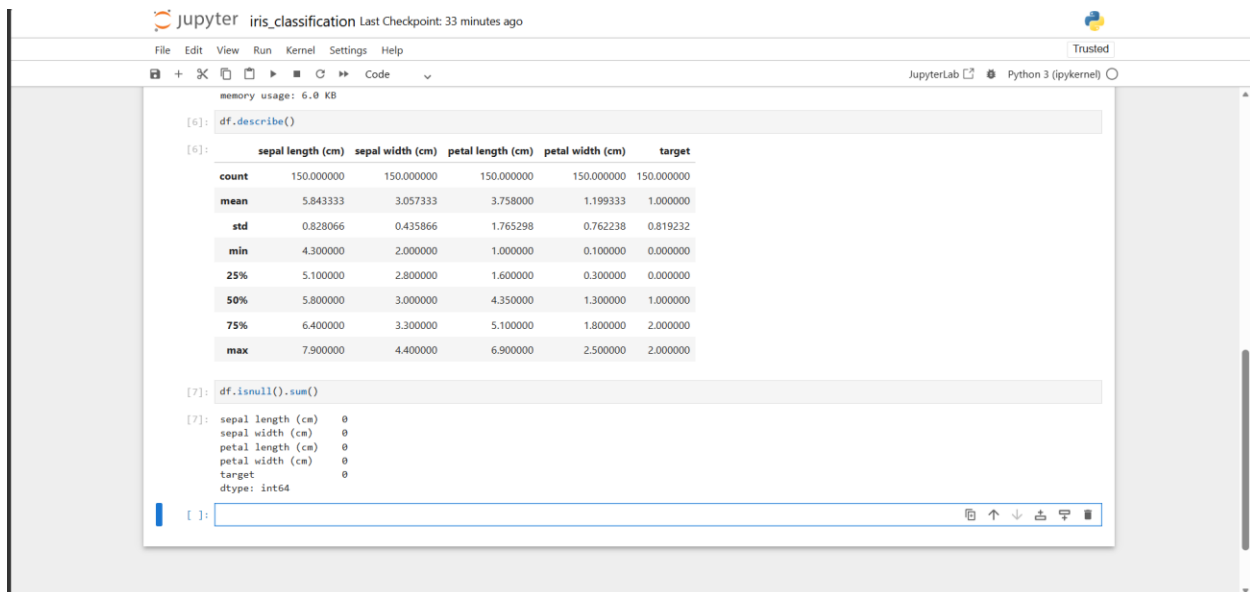
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   sepal length (cm)      150 non-null   float64
1   sepal width (cm)       150 non-null   float64
2   petal length (cm)      150 non-null   float64
3   petal width (cm)       150 non-null   float64
4   target                 150 non-null   int64   
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

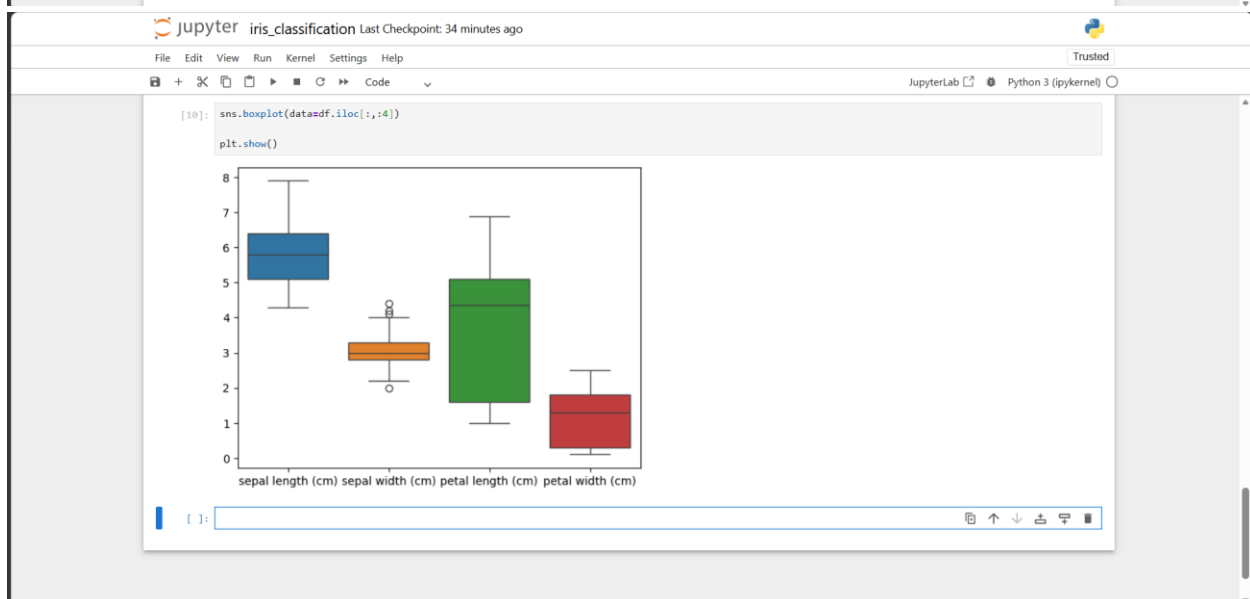
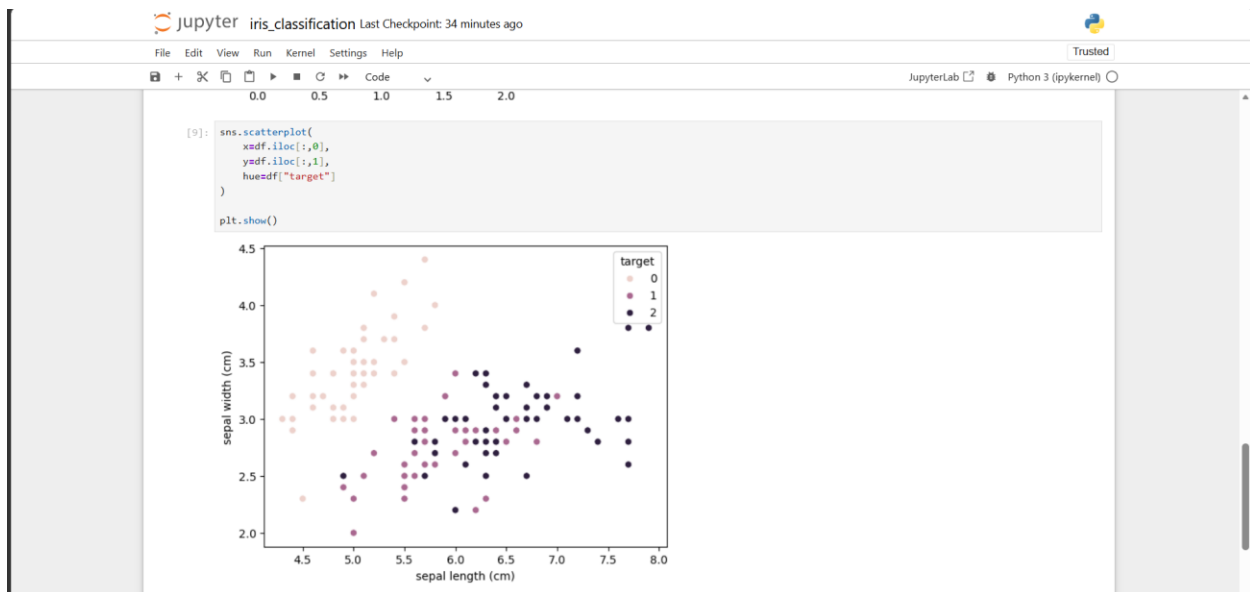
```
Data columns (total 5 columns):
# Column Non-Null Count Dtype
---
0 sepal length (cm) 150 non-null float64
1 sepal width (cm) 150 non-null float64
2 petal length (cm) 150 non-null float64
3 petal width (cm) 150 non-null float64
4 target 150 non-null int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

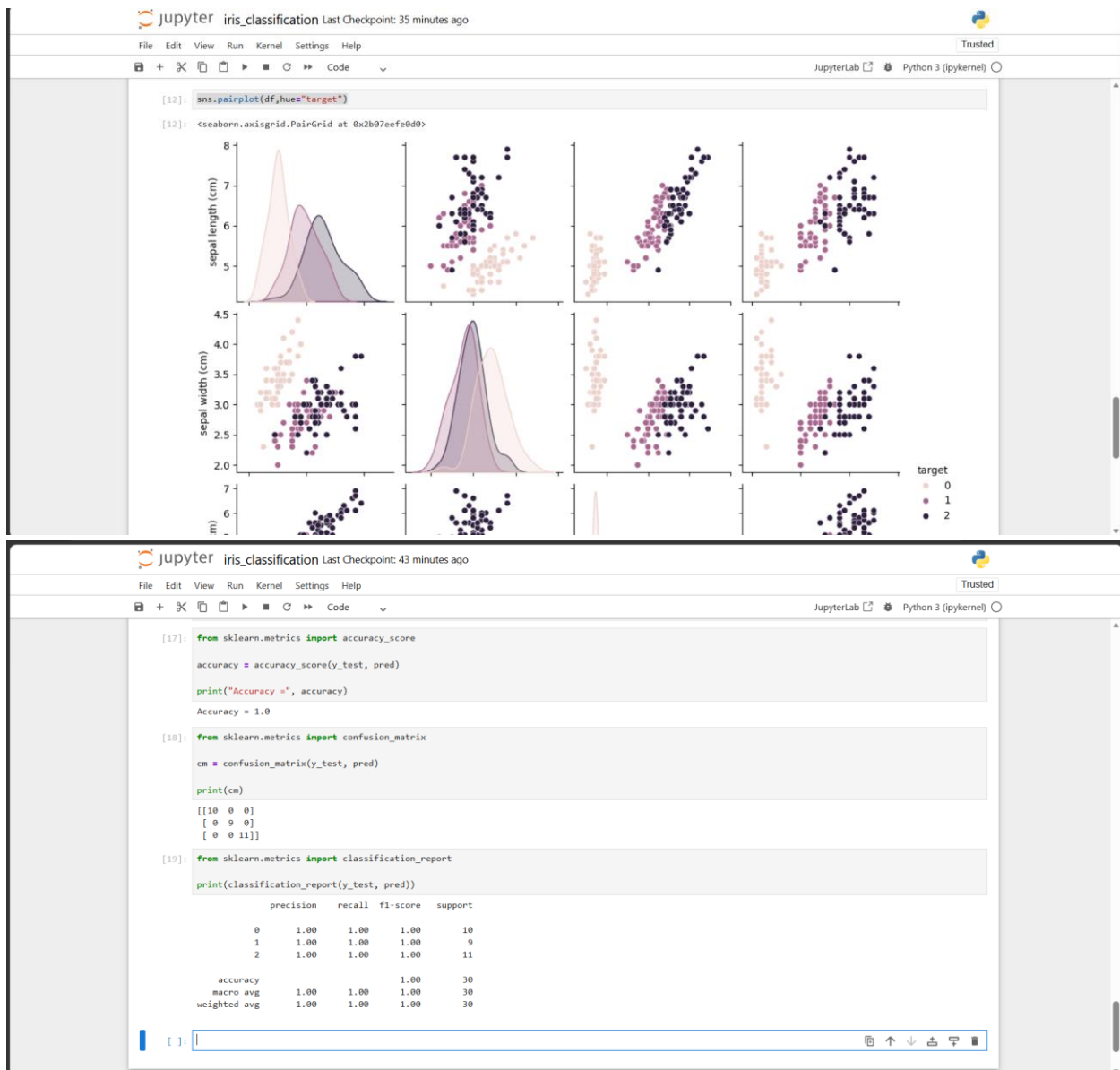
```
[6]: df.describe()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

```
[ ]:
```







5. Data Visualization

Several visualizations were created to understand the dataset.

Visualizations Performed

1. Histogram
2. Scatter Plot
3. Box Plot
4. Pair Plot

Observations

- Petal features provide strong separation among classes.
- Most features follow distinguishable distributions.
- Some outliers are visible in the box plot.
- Pair plots clearly show class separation.

Screenshots

[Insert Histogram Screenshot]

[Insert Scatter Plot Screenshot]

[Insert Box Plot Screenshot]

[Insert Pair Plot Screenshot]

6. Model Building

Algorithm Used

Logistic Regression

Steps Performed

1. Feature and target selection
2. Train-test split
3. Model training
4. Prediction on test data
5. Model evaluation

7. Model Evaluation

Metrics Used

- Accuracy Score
- Confusion Matrix
- Classification Report

Results

Accuracy Achieved: _____

(Write the accuracy value obtained from your notebook)

Screenshots

[Insert Accuracy Screenshot]

[Insert Confusion Matrix Screenshot]

[Insert Classification Report Screenshot]

8. Model Saving

The trained model was saved using Joblib.

File Name:

iris_model.pkl

This model can be loaded later and used for prediction without retraining.

9. Conclusion

The Iris Classification project was successfully completed using Logistic Regression. The dataset was explored, visualized, and used to train a machine learning model. The model achieved high classification accuracy and demonstrated the effectiveness of machine learning techniques for classification problems.

This project helped in understanding the complete machine learning workflow including data preprocessing, visualization, model training, evaluation, and model deployment preparation.

10. Repository Artifacts

Files Included:

- iris_classification.ipynb
- iris_model.pkl
- iris_report.pdf
- Screenshots

Submitted as part of the InternSpark Machine Learning Internship Program.

